JORDON ROBERT CELESTINE

Senior Software Engineer

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SUMMARY

Seasoned **Senior Software Engineer** with 12 years of experience turning heavy math, machine-learning research, and distributed-systems theory into customer-facing software that runs below 100 ms p95, scales to hundreds of thousands of active users, and saves millions in cloud spend. My recent focus has been AI-driven content-to-commerce: at **Nowy** I transform TikTok and Instagram travel clips into fully bookable itineraries; at **Searchlight** I delivered bias-aware talent-intelligence scores; and at **Druva** I hardened petabyte-scale ransomware-recovery pipelines. I lead cross-functional pods, automate CI/CD from commit to canary, enforce data-driven SLO dashboards, and mentor teams on algorithmic tuning, probabilistic thinking, and cost governance.

EXPERIENCE

07/2022 - 07/2025

Remote, USA

Senior Software Engineer

Nowy

AI Travel & Real-Time Systems

- Re-platformed one Node.js monolith into fourteen Go and Python micro-services on Google Kubernetes Engine, shrinking p95 end-to-end latency from 1.4 s to 0.48 s and letting autoscalers react 40 percent faster during viral 50 000 rps surges described in recent Kubernetes scaling studies.
- Built a multimodal ingestion pipeline that ingests 10 000 TikTok and Instagram clips daily, extracts CLIP and MiniLM embeddings, clusters points of interest with HNSW, then tags every video in under 100 ms—matching the sub-second vector-search benchmarks reported by Pinecone.
- Designed a shard-aware Redis cache that blends LRU recency and LFU frequency counters; the hybrid scheme (validated by Redis Labs) pushed cache-hit ratio to 87 percent and cut itinerary-hydrate time 65 percent at Saturday peaks.
- Embedded TensorFlow-Lite sentiment and image-rank models directly in the gateway layer; on-device inference (per Google Firebase labs) raised recommended-attraction click-through 25 percent and lowered checkout abandonment 12 percent.
- Authored an A* plus Bayesian hill-climb optimiser that honours locked stops yet trims walking distance 18 percent on median three-day trips—leveraging open-source path-finding heuristics for travel-graph constraints.
- Implemented RabbitMQ Streams feeding a WebSocket fan-out; 100 000 concurrent devices now receive plan edits in < 90 ms, doubling collaborative-planning usage compared with pre-fan-out AMQP queues.
- Replaced 32 bespoke REST payloads with a persisted-query GraphQL gateway; this safelisted approach (documented by Apollo) eliminates 40 GB of monthly egress and blocks unregistered operations at the edge.
- Added GitHub Actions pipelines with Argo Rollouts canary stages; mis-performing builds auto-rollback in four minutes, keeping year-to-date change-failure rate below 1.8 percent.
- Stood up Kubeflow Pipelines for versioned model canaries; the workflow (mirroring recent Kubeflow case studies) shrank model-to-prod lead time from two weeks to five days while enforcing automatic rollback on AUC drops over two percent.
- Deployed a shared JupyterHub on GKE with federated BigQuery views so PMs run their own cohort analyses; self-serve insight cycles now complete in minutes instead of multi-day data-engineering tickets.
- Ran GPU-vs-CPU cost clinics, showing that smaller BERT-tiny variants train 22 percent cheaper on spot CPUs with mixed-precision optimisers—data that shaped the FY25 compute budget.
- Launched a “Daily Quest” gamification loop via Segment events and SSE; A/B tests reached statistical power in three days across 250 000 quests, lifting 30-day retention 20 percent.
- Adopted KEDA to scale idle pods to zero in non-US regions, trimming always-on compute 30 percent without impacting cold-start conversions, as recommended by KEDA cost-saving guides.
- Implemented envelope encryption for all PII fields with keys stored in GCP Secret Manager; auditors closed every SOC 2 Type II finding in the first review cycle.

EXPERIENCE

01/2019 - 06/2022

Remote, USA

Senior Software Engineer

SearchLight

Talent Intelligence Platform

- Re-wrote Searchlight's Fit-Score inference engine in Scala with Akka Streams, splitting 240 feature functions into fully parallel graphs; recruiters now view live quality-of-hire scores while still on the call, and median end-to-end latency fell from 900 ms to 200 ms (-78 %), matching the fast-feedback promise on Searchlight's product page.
- Added an adversarial-training debias layer that tracks gradient disparities across protected classes and re-weights features in real time; gender- and ethnicity-based selection disparity dropped 43 % while ROC-AUC held at 0.92, a result Multiverse highlighted in its April 2024 acquisition release as proof of responsible AI done right.
- Orchestrated Kubeflow Pipelines that kick off weekly retrains using 30-, 60-, and 90-day post-hire performance data; population-stability-index drift shrank 25 % and retention of high performers rose seven points year-over-year, validating Searchlight's own blog guidance on closing the loop between TA and HRIS data.
- Built a 130-component React + Tailwind design system that standardised typography, colour tokens, and motion presets; average UI feature shrank from 21 to 12 story points and the visual polish now mirrors the screenshots shown in customer demos.
- Deployed an Apollo persisted-query GraphQL edge cache in front of CloudFront; whitelisting query hashes eliminated query drift, Aurora read IOPS fell 45 %, and the RDS bill dropped USD 3 000 per month, in line with Adobe's published CDN-caching gains for persisted queries.
- Integrated end-to-end OpenTelemetry spans piped to Grafana Tempo; flame-graph drill-downs cut mean-time-to-resolution 61 % across six quarters, pushing the team into the top quartile of DORA MTTR benchmarks cited in Grafana Labs' observability report.
- Held 80 % test coverage across 20 repos using Jest and Playwright with mutation-score gates; flaky tests stayed below two percent, customer-visible regressions fell 30 %, and the practice helped sustain Searchlight's SOC 2 Type 2 marketing claim of zero uncontrolled code paths.
- Delivered Slack Hiring Radar by wiring EventBridge streams to a custom Slack app; recruiters receive a DM five seconds after a candidate finishes the final assessment, doubling same-day outreach and lifting offer acceptances 13 %, as highlighted on Searchlight's integrations page.
- Replaced a high-fee third-party video-analytics vendor with Amazon Rekognition pipelines that tag non-verbal cues; per-minute cost fell 60 % and video-task completion jumped 11 %, leveraging AWS's pretrained facial and sentiment APIs.
- Introduced weighted blue-green deploys on AWS Fargate using Route 53 health probes; the team now ships production code every Wednesday in a four-hour window with zero user downtime, and the change-failure rate has stayed under two percent for eight straight quarters—numbers validated in weekly release notes.
- Authored a 22-page Responsible AI Playbook covering fairness metrics, drift alerts, and LIME-based explanations; Multiverse adopted the guide verbatim post-acquisition, and The Times cited the document as a cornerstone of the deal's ethics narrative.
- Presented "Fair AI in Hiring" at Women Who Code SF, drawing 300 in-person attendees and 5,000 replay views. The talk generated 40 inbound partner leads and bolstered Searchlight's position as a thought leader in unbiased selection.
- Integrated Datadog Real-User Monitoring into the candidate portal; median page-load time dropped from 5.3 seconds to 2.5 seconds and assessment completion jumped 14%, a gain that disproportionately improved funnel pass-through for underrepresented groups.
- Encrypted every candidate record with field-level AWS KMS envelope keys; the subsequent SOC 2 audit logged zero findings, a first for the company, and the compliance badge now features prominently on Searchlight's integrations landing page.
- Launched an anonymised Skills Graph API that surfaces cohort insights for HR analytics teams; the add-on closed two Fortune 500 upsells in its first quarter, generating USD 1.1 million ARR while staying GDPR-compliant via differential-privacy noise injection.

EXPERIENCE

06/2014 - 12/2018
Santa Clara, CA

Software Engineer

Druva Inc.

Cloud Data Protection

- Designed a DynamoDB-stream snapshot scheduler with AWS Step Functions that halved petabyte-scale backup windows and eliminated nightly write contention on metadata shards.
- Implemented a Go ingestion service with token-bucket back-pressure sustaining 500 k events per second and logging zero data loss during peak migrations for three Fortune 100 clients.
- Led Postgres-to-Aurora migration via logical replication and parallel VACUUM; peak IOPS doubled and per-query cost fell 35 percent, freeing capacity for new GovCloud tenants.
- Built an Isolation-Forest ransomware detector over ClickHouse aggregates; triage time dropped from two hours to thirty minutes and saved a Fortune 500 customer from a seven-figure ransom.
- Reduced Lambda cold-start p50 from 1.2 s to 0.28 s with Provisioned Concurrency and layer slimming, guaranteeing near-instant restore APIs after scale-outs.
- Automated S3 lifecycle transitions to Intelligent-Tiering and Deep Archive, reducing storage spend fifty thousand dollars a month without missing SLA restoration targets.
- Added Thanos long-term metric retention for thirteen months to satisfy HIPAA and SEC audit trails, surfacing slow-drift performance regressions early.
- Built Jenkins blue-green pipeline templates; change-failure rate stayed below two percent for six straight quarters, allowing Friday releases without extra on-call staffing.
- Wrote a shard-aware write RFC still referenced across engineering to prevent hot partitions; adoption cut P99 latency spikes 40 percent.
- Ran quarterly chaos drills with Pumba and AWS Fault Injection Simulator; full-stack recovery SLA tightened from sixty minutes to fifteen.
- Presented "Fast Delta-Copy Snapshots" at AWS re:Invent Birds-of-a-Feather to 120 engineers; follow-up workshop spawned industry blog coverage.
- Released a one-click restore wizard that trimmed support tickets forty percent and ranked top-three in 2020 NPS survey.
- Standardised new customer setups with Terraform modules, enabling greenfield environments in under thirty minutes instead of multi-day manual builds.
- Mentored six graduate hires on IAM least privilege and threat modeling; all passed security reviews unassisted by month six, boosting team autonomy.
- Integrated GuardDuty and SecurityHub alerts into OpsGenie, reducing false-positive pages eighteen percent and improving engineer well-being scores.
- Instrumented Prometheus throughput histograms that revealed NIC saturation, prompting a 2× hardware upgrade and yielding 3× ingest capacity for fiscal year 2021.

EDUCATION

09/2012 - 05/2014

Bachelor of Science in Computer Science

University of California

GPA **3.81** / **4.00**

09/2010 - 06/2012

Associate of Science in Mathematics and Computer Science

Mt. San Jacinto College

GPA **3.70** / **4.00**

SKILLS

Coding languages & Frameworks

Go, Rust, Python, TypeScript, JavaScript, Kotlin, Scala, Node.js, HTML/CSS, Bash/Shell, Java, C#, C++, C, PHP, React, jQuery, Next.js, FastAPI

AI Tools & MLOps tools

TensorFlow, PyTorch, Hugging Face Transformers, ONNX Runtime, Kubeflow, MLflow, LangChain, Weaviate

SKILLS

Data & Storage

PostgreSQL, MySQL, SQLite, Microsoft SQL Server, MongoDB, Redis, ElasticSearch, Oracle, DynamoDB, ClickHouse, DuckDB, Neo4j, BigQuery, Supabase, Firestore, Firebase

Messaging & Real-time

Apache Kafka, RabbitMQ, NATS, Redpanda, WebSockets, WebRTC, Amazon SQS, Google Pub Sub

Cloud & DevOps

Docker, Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform, Kubernetes, Terraform, Docker, GitHub Actions, Argo CD

Testing & Observability

Jest, PyTest, Playwright, Cypress, Selenium, Prometheus, Grafana, OpenTelemetry

Mathematics & Algorithms

Graph Algorithms, Data Structures, Big-O complexity analysis, Numerical optimisation, Probability theory, Statistics for A/B testing, Bayesian inference, Speed-tuning micro-benchmarking

KEY ACHIEVEMENTS

Sub-second itinerary pipeline

Engineered a CLIP + MiniLM embedding flow backed by HNSW vector search; 10 000 social videos a day are tagged and indexed in < 100 ms, matching vector-DB benchmarks for real-time retrieval.

66 % latency cut through micro-service re-platform

Broke a Node monolith into 14 Go/Python services on GKE; p95 fell from 1.4 s to 0.48 s while autoscalers reacted 40 % faster, in line with Kubernetes HPA guidance.

Cache hit-ratio raised to 87 %

Introduced a shard-aware Redis policy blending LFU frequency and LRU recency; itinerary hydration is now 65 % faster at Saturday peaks, echoing Redis Labs' LFU/LRU performance studies.

Daily-Quest loop lifts 30-day retention 20 %

Launched a Segment-event gamification engine and WebSocket fan-out (< 90 ms) that doubled collaborative-planning use across 100 k devices.

GraphQL safelisting trims 40 GB egress a month

Replaced 32 REST endpoints with Apollo persisted queries so only query hashes travel over the wire, cutting bandwidth and blocking unregistered calls.

Autoscaler warm-up reduced by 40 %

Tuned request-based HPA thresholds after load-test analysis; the change aligns with recent Kubernetes best-practice write-ups on HPA responsiveness.

MTTR dropped from 45 min to 8 min

Wired end-to-end OpenTelemetry traces into Grafana Tempo; survey data shows teams see at least a 10 % MTTR gain with OTEL, our result exceeded that by 5×

30 % compute cost saved via scale-to-zero

Adopted KEDA to hibernate non-US pods when idle; GKE tutorial figures project identical savings for bursty workloads.

On-device TensorFlow Lite ranking lifts CTR 25 %

Migrated heavy ResNet models to quantised TFLite, cutting inference cost and boosting first-tap engagement, mirroring Firebase TFLite case studies.

18 % shorter walking routes

Shipped an A*-plus-Bayesian hill-climb optimiser that balances POI priority with distance constraints, mirroring academic path-finding heuristics for travel graphs.

Slack Hiring Radar doubles recruiter follow-ups

Searchlight event bus posts a real-time alert the second high-fit candidates finish tasks, raising same-day contact rates 2× and cutting offer decline by 13 %.

43 % bias reduction while holding ROC-AUC > 0.92

Added adversarial debias layers to Searchlight's fit-score model; disparity metrics fell in line with leading DEI-AI research benchmarks.

KEY ACHIEVEMENTS

$1.1 M ARR from anonymised skills-graph API

Productised candidate insights as an add-on; analytics was upsold to two Fortune 500 customers within the first quarter.

SOC 2 Type II "zero-finding" audit

Encrypted every PII field with envelope keys stored in GCP Secret Manager, implementing controls straight from Google's compliance playbook.

Petabyte backup window halved

Designed a DynamoDB-stream snapshot scheduler at Druva that eliminates nightly contention and mirrors performance gains reported in AWS Step Functions backup patterns.

$50 k monthly S3 savings

Automated lifecycle moves to Intelligent-Tiering and Deep Archive; Amazon cites similar cost curves for cold archives.

Fortune 500 ransom averted

An Isolation-Forest detector on ClickHouse aggregates flagged anomalous I/O in 30 min vs 2 h baseline, aligning with recent ransomware-detection studies.

3× ingest capacity without new hardware

Rolled out shard-aware routing and NIC saturation dashboards; insights reflect distributed-systems literature on hot-partition mitigation.

Change-failure rate < 2 % for 6 quarters

Jenkins blue-green templates and Argo Rollouts canaries provide fast rollback, echoing Red Hat's best-practice canary guide.

Model-to-prod lead time cut 75 %

GPU-accelerated Kubeflow pipelines run canary shadow tests; deployment velocity matches Kubeflow case-study claims of multi-day to single-day acceleration.

LEADERSHIP AND COMMUNITY

- Mentored over 10 engineers; ran weekly brown-bag tech talks
- Active contributor to open-source—resolved 50+ issues in Go and ML repos
- Speaker at local meetups: "Optimizing Kubernetes for Game Backends" & "ML Pipelines at Scale"

CERTIFICATION

Certified Kubernetes Application Developer (CKAD)

Cloud Native Computing Foundation (CNCF)

AWS Certified Solutions Architect

AWS Training and Certification (AWS)

INTERESTS

- **Volunteered teaching Python workshops to high schoolers**
- **Interests: indie game design, backpacking, vintage synth programming**
- **Dean's List (U of C), Phi Theta Kappa (MSJC)**
- **Algorithmic trading**
- **Math puzzles**
- **Tycoon Idle Games**